

AUTOMATIC HEIGHT MEASURING DEVICE

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- **AIM OF THE PROJECT:** –

To measure the height of a person accurately without any hassle.

- **THEORY BEHIND THE PROJECT:** –

It uses **high frequency sound waves** (ultrasonic waves) to resonate a desired frequency and convert electric energy into acoustic energy, and vice versa. Sound waves are transmitted to and reflected from the target back to the transducer.

- **MATERIALS REQUIRED:** –

(i) 1x Arduino Uno Board

(ii) 1x Ultrasonic Sensor

(iii) 1x Liquid Crystal Display (LCD) with Inter-Integrated Circuit (I²C)

(iv) 2x Light Emitting Diodes (LED)

(v) 12x Jumper Wires

- **WORKING:** –

The ***Ultra Sonic Sensor*** transmits high frequency sound waves. These sound waves when are reflected of a person's head it measures the distance between it and person's head by simply multiplying speed of sound, i.e. around 343m/s, by time taken. Then it finds out the height of the person by subtracting the distance between it and person's head from the height it is placed, which is 200 cm in this scenario.

- **ADVANTAGES:** –

(i) It is more accurate than most of the traditional methods. (Under special circumstances)

(ii) It is less time consuming and hassle free.

(iii) It can be accurate up to 8 decimal places. (Under special circumstances)

(iv) It can save time in mass health check-ups and measurements.

(v) Can be used by elderly person easily.

- **APPLICATIONS:** –

(i) Can be used in Medical Centres and Hospitals to save time during community check-ups.

(ii) Can be used in old age homes and orphanages for convenience.

(iii) When calibrated, it can be used to measure small objects.